

CCNA 200-301 v1.1 - Last Minute Review (10 Pages)

This review follows the official CCNA exam blueprint topics and is intended as a concise memory refresher before the exam. The blueprint itself provides objectives rather than IOS command references; the commands included here are representative study commands aligned to those objectives. ■filecite■turn2file0■

1. Network Fundamentals

- Know: Routers, L2/L3 switches, NGFW, APs, controllers, servers, endpoints, PoE.
- TCP vs UDP, IPv4 vs IPv6, private IPv4 ranges, IPv6 address types, subnetting, MAC learning, ARP.
- Commands: show ip interface brief, show ipv6 interface brief, show arp, show mac address-table, ping, traceroute.
- Exam traps: Longest Prefix Match determines the best route before Administrative Distance.

Review Tip: Focus on understanding what each command verifies, not just its syntax.

2. VLANs & Trunking

- Commands: `vlan` , `name NAME`, `switchport mode access`, `switchport access vlan X`, `switchport mode trunk`, `switchport trunk native vlan X`.
- Verify: `show vlan brief`, `show interfaces trunk`, `show interfaces switchport`.
- Remember: VLAN 1 = default VLAN. Native VLAN carries untagged traffic.

Review Tip: Focus on understanding what each command verifies, not just its syntax.

3. EtherChannel & STP

- LACP: channel-group X mode active/passive.
- Verify: show etherchannel summary, show spanning-tree.
- Remember: PortFast = immediate forwarding. BPDU Guard = err-disable on BPDU. BPDU Filter = filters BPDUs. Root Guard prevents another root. Loop Guard protects alternate ports.

Review Tip: Focus on understanding what each command verifies, not just its syntax.

4. Routing & OSPF

- Commands: ip route, ipv6 route, router ospf 1, network A.B.C.D W area 0, router-id, passive-interface.
- Verify: show ip route, show ip ospf neighbor, show ip ospf interface brief.
- Exam traps: Longest Prefix Match, Administrative Distance, Metric.

Review Tip: Focus on understanding what each command verifies, not just its syntax.

5. IP Services

- NAT, DHCP, DNS, SSH, NTP, SNMP, Syslog, QoS.
- SSH: hostname, ip domain-name, crypto key generate rsa, username, line vty, login local, transport input ssh.
- Verify: show ip nat translations, show ip ssh, show ntp associations.

Review Tip: Focus on understanding what each command verifies, not just its syntax.

6. Security

- ACLs: access-list / ip access-list, permit, deny, ip access-group.
- Port Security: switchport port-security, maximum, mac-address sticky.
- Know DHCP Snooping, DAI, AAA, VPN concepts, WPA2/WPA3.

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7. Automation

- Know JSON, REST APIs, SDN, Controllers, Ansible, Terraform.
- HTTP verbs: GET, POST, PUT, PATCH, DELETE.
- Know CRUD mapping and northbound/southbound APIs.

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8. High-Yield Show Commands

- show running-config
- show startup-config
- show ip interface brief
- show interfaces
- show vlan brief
- show interfaces trunk
- show spanning-tree
- show etherchannel summary
- show ip route
- show access-lists
- show mac address-table
- show arp

Review Tip: Focus on understanding what each command verifies, not just its syntax.

9. High-Yield Exam Traps

- PortFast $\neq$ BPDU Guard.
- BPDU Filter does NOT err-disable a port.
- One Root Port per non-root switch.
- Native VLAN traffic is untagged.
- Wildcard masks are inverse subnet masks.
- Default Administrative Distances: Connected 0, Static 1, OSPF 110.

Review Tip: Focus on understanding what each command verifies, not just its syntax.

10. Final Memory Sheet

- Memorize: subnetting, VLAN/trunk commands, EtherChannel, STP roles, static routes, OSPF basics, NAT, SSH, ACL syntax, Port Security, DHCP Snooping, DAI, REST verbs.
- Practice every configuration in Packet Tracer and always verify with show commands.

Review Tip: Focus on understanding what each command verifies, not just its syntax.